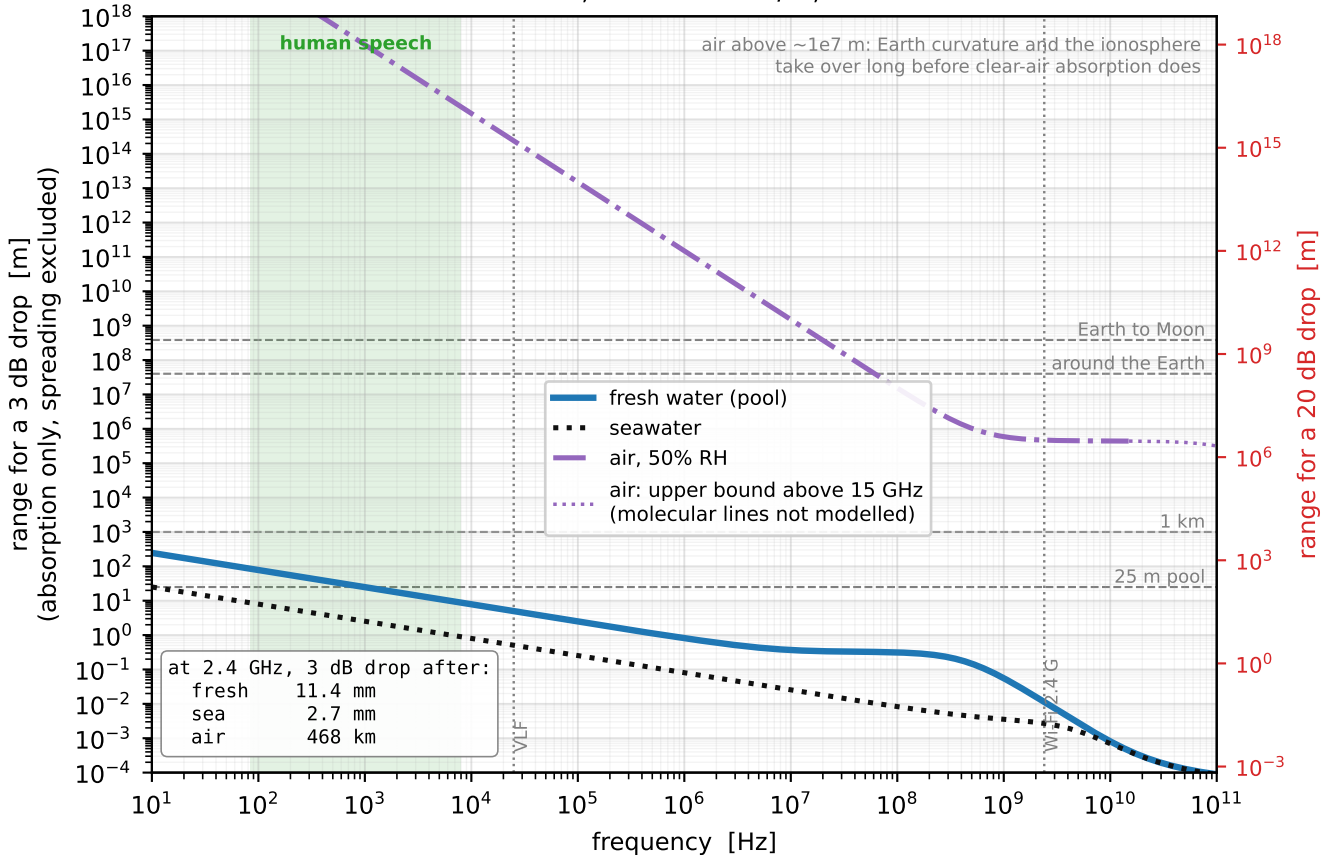
Electromagnetic waves: how far before the medium takes 3 dB
fresh σ = 0.05, sea σ = 4.8 S/m, air 50% RH
human speech
air above ~1e7 m: Earth curvature and the ionosphere
take over long before clear-air absorption does
Earth to Moon
around the Earth
1 km
25 m pool
fresh water (pool)
seawater
air, 50% RH
air: upper bound above 15 GHz
(molecular lines not modelled)
at 2.4 GHz, 3 dB drop after:
fresh 11.4 mm
sea 2.7 mm
air 468 km
VLF
Wi-Fi 2.4 G
range for a 3 dB drop [m]
(absorption only, spreading excluded)
range for a 20 dB drop [m]
frequency [Hz]